

Vertebroplasty and Kyphoplasty

ACG: A-0226 (AC)

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Clinical Indications for Procedure

- Vertebral augmentation via percutaneous vertebroplasty or kyphoplasty may be indicated for **ALL** of the following(1)(2)(3)(4)(5)(6)(7)(8):
 - Clinical diagnosis of **1 or more** of the following:
 - Osteoporotic vertebral compression fractures(11)(12)(13)[N](#)
 - Malignancy-related vertebral compression fractures(11)(31)[N](#)
 - Thoracic or lumbar vertebral fracture confirmed by advanced imaging (ie, MRI with bone marrow edema, bone scan, or SPECT-CT)
 - Failure of nonsurgical management (eg, oral analgesics, rest, bracing, physical therapy)
 - Pain causing loss of mobility or inability to perform baseline level of daily activities
 - No osteomyelitis at surgical site
 - No unstable vertebral fracture

Alternatives to Procedure

- Alternatives include:
 - For bone metastasis(35)(36)(37):
 - Analgesics(38)
 - Radiation therapy, including stereotactic body radiotherapy. See Stereotactic Body Radiotherapy [AC](#) for further information.
 - Radiofrequency ablation. See Radiofrequency Ablation of Tumor [AC](#) for further information.
 - For osteoporotic fracture(39)(40):
 - Activity modification
 - Analgesics(38)
 - Bracing(41)(42)
 - Calcitonin(43)
 - Physical therapy

Evidence Summary

Background

Minimally invasive, image-guided, percutaneous vertebral body augmentation techniques, such as vertebroplasty and kyphoplasty, are options for the management of severe pain associated with vertebral compression fractures secondary to osteoporosis or malignancy. (1)(3)(8)(9) **(EG 2)**

Vertebroplasty is a technique by which polymethyl methacrylate bone cement is injected through a large-bore access needle into a fractured vertebral body allowing for stabilization and restoration of bone strength. Kyphoplasty involves the insertion of an inflatable balloon into the vertebral body to create a cavity within the bone followed by injection of polymethyl methacrylate bone cement to fill the space.(4)(8)(9)(10) **(EG 2)**

Criteria

The evidence for the clinical indications found in this guideline includes 28 published peer reviewed articles and 3 specialty society or other evidence-based guidelines.

For osteoporotic vertebral compression fractures, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A systematic review and meta-analysis of 8 studies evaluating the efficacy of vertebral augmentation (percutaneous vertebroplasty with spinal implant as compared with balloon kyphoplasty) in 1027 patients with osteoporotic vertebral compression fractures found that percutaneous vertebroplasty with spinal implant was associated with improved postoperative pain and function (as measured by visual analog scale and Oswestry Disability Index, respectively) as well as reduced cement extravasation as compared with kyphoplasty; there was no difference in postoperative quality of life (as measured by EuroQol 5-Domain Questionnaire (EQ-5D)) between groups. The authors recommended further high-quality randomized controlled trials.(14) **(EG 1)** A systematic review with meta-analysis identified 7 studies including more than 2 million patients with osteoporotic vertebral compression fractures treated with either vertebral augmentation (vertebroplasty or balloon kyphoplasty) or nonsurgical management and found that vertebral augmentation was associated with a reduction in mortality risk at 2-year and 5-year follow-up (30% and 21%, respectively) as compared with conservative management. Additionally, balloon kyphoplasty provided mortality benefits over vertebroplasty.(15) **(EG 1)** A systematic review with meta-analysis including 12 randomized controlled trials (1110 participants) examined if percutaneous vertebral augmentation for osteoporotic vertebral compression fractures resulted in better improvement in pain when patients were selected based on clinical examination plus imaging or clinical examination alone and found that randomized controlled trials that used imaging to confirm bone marrow edema in the suspect vertebra showed an improved size effect as compared with randomized controlled trials using no imaging. The authors reported that this benefit was sustained through the immediate, short, intermediate, and long term.(16) **(EG 2)** A randomized controlled trial including 180 patients with acute osteoporotic vertebral compression fractures compared treatment with either vertebroplasty or a sham procedure (local subcutaneous lidocaine and vertebral pedicle periosteum infiltration with bupivacaine) and found that both vertebroplasty and the sham procedure were associated with pain reduction (measured with a visual analog scale) at all follow-up points through 12 months compared with baseline; however, vertebroplasty was not associated with greater pain relief compared with the sham procedure. The authors noted that the study was limited by potential selection bias due to nearly half of eligible patients declining study enrollment and the potential for the sham periosteal infiltration alone to provide enough pain relief in the acute phase without the need for additional cementation.(17) **(EG 1)** A randomized controlled trial including 52 patients with acute osteoporotic vertebral compression fractures (less than or equal to 8 weeks' duration) compared treatment with either vertebroplasty or a sham procedure (lidocaine injected into fractured vertebral body) and found that both vertebroplasty and the sham procedure were associated with pain reduction (measured with a visual analog scale) as compared with baseline; however, the patients in the sham procedure group had significantly higher pain scores with movement throughout the follow-up period. The authors noted that the study was limited by the small sample size due to difficulty with recruitment and a low inclusion rate.(18) **(EG 1)** A randomized controlled trial including 135 patients with acute osteoporotic vertebral compression fractures compared treatment with either vertebroplasty or conservative therapy and found that vertebroplasty was associated with a greater reduction in pain (measured with a visual analog scale) and disability (measured by Oswestry Disability Index) and improved quality of life (measured by Quality of Life Questionnaire of the European Foundation for Osteoporosis) at all time points through 1-year follow-up as compared with conservative management, though both treatment groups reported significant improvement in these measures from baseline. The authors noted that the study was limited by the unmasked design, single-center location, and the small sample size; a multicenter study with longer-term follow-up was planned.(19) **(EG 1)** A randomized controlled trial including 120 patients with severe pain (greater than or equal to 7 on a numeric rating scale) due to an acute osteoporotic vertebral compression fracture (less than or equal to 6 weeks' duration) compared treatment with either vertebroplasty or a sham procedure (subcutaneous lidocaine) and found, at all follow-up times through 6 months, that vertebroplasty was associated with a greater proportion of patients achieving a pain score of less than or equal to 4 and a greater mean reduction in pain scores as compared with the sham procedure.(20) **(EG 1)** An open-label randomized controlled trial including 82 patients with acute osteoporotic vertebral compression fractures compared treatment with either vertebroplasty or optimal medical therapy and found that vertebroplasty was associated with improvement in pain (measured with a visual analog scale) and quality of life (measured with a questionnaire based on the Oswestry Low Back Pain Disability Questionnaire) at 24-month follow-up, and restoration of vertebral bone height and prevention of spinal deformity at 36-month follow-up, compared with optimal medical therapy.(21) **(EG 1)** An open-label randomized controlled trial including 202 patients with acute osteoporotic vertebral compression fractures (less than or equal to 6 weeks' duration) compared treatment with either vertebroplasty or conservative management and found, at 1-month and 1-year follow-up, that vertebroplasty was associated with a greater reduction in pain scores from baseline (measured with a visual analog scale) as compared with conservative management. The authors noted that the study was limited by its open-label design, which may have introduced bias.(22) **(EG 1)** An open-label randomized controlled trial including 50 patients with acute and subacute osteoporotic vertebral fractures compared treatment with either vertebroplasty or conservative treatment and found that vertebroplasty was associated with a reduction in pain 12 to 24 hours after the procedure (measured by a visual analog scale); however, at 3-month and 12-month follow-up, both vertebroplasty

and conservative treatment were associated with a reduction in pain scores without a significant between-group difference. The authors noted that the study was limited by incomplete baseline pain assessment and small sample size.(23)(24) **(EG 1)** A randomized controlled trial including 131 patients with painful osteoporotic vertebral compression fractures (up to 1-year duration) compared treatment with either vertebroplasty or a sham procedure (local subcutaneous lidocaine and vertebral pedicle periosteum infiltration with bupivacaine) and found, through 1-month follow-up, that both vertebroplasty and the sham procedure were associated with an immediate reduction in disability (measured by Roland-Morris Disability Questionnaire) and pain (based on a numeric rating scale) compared with baseline; however, vertebroplasty was not associated with a greater reduction in disability or pain compared with the sham procedure. The authors noted that the study was limited by the allowance of crossover at 1 month, thereby affecting the ability to perform longer-term follow-up between treatment groups.(25) **(EG 1)** A randomized controlled trial including 78 patients with painful osteoporotic vertebral compression fractures (up to 1-year duration) compared treatment with either vertebroplasty or a sham procedure (local and posterior laminal periosteum anesthetic injection) and found that both vertebroplasty and the sham procedure were associated with a reduction in overall pain at all follow-up time points; however, vertebroplasty was not associated with greater pain relief compared with the sham procedure. The authors noted that the study was limited by difficulty in recruitment, the small sample size, and potential selection bias, with nearly 1/3 of eligible patients declining study participation.(26) **(EG 1)** A randomized controlled trial including 300 patients with painful vertebral compression fracture compared treatment with either balloon kyphoplasty or nonsurgical care and found that balloon kyphoplasty was associated with improvement in pain score and disability score (measured by the Roland-Morris Disability Questionnaire) through 12 months and improvement in quality-of-life measures through 6 months as compared with conservative management. The authors noted that the study was limited by the open-label design, a high dropout rate in the control group, and nonstandardized nonsurgical intervention.(27) **(EG 1)** A prospective phase IV clinical trial including 354 patients with painful acute or subacute vertebral compression fractures associated with osteoporosis or cancer evaluated the benefit of balloon kyphoplasty and found, at 3-month follow-up, that kyphoplasty was associated with an improvement in pain (measured with a numeric rating scale), disability (measured by Oswestry Disability Index), and quality of life (measured by physical component summary of the 36-Item Short Form Health Survey and EQ-5D) as compared with baseline. The authors noted that the study was limited by its open-label nonrandomized design.(28) **(EG 2)** A national practice guideline recommends percutaneous vertebroplasty and percutaneous balloon kyphoplasty without stenting as options for treatment of osteoporotic vertebral compression fractures in patients with severe ongoing pain at the level of an acute unhealed vertebral fracture confirmed by physical examination and imaging despite optimal pain management (ie, pain medication, nonsurgical).(29) **(EG 2)** While one specialty society clinical practice guideline recommends against the use of vertebroplasty for osteoporotic spinal compression fractures,(30) multiple specialty societies and an expert consensus panel recommend percutaneous vertebral augmentation (vertebroplasty or kyphoplasty) for the treatment of painful vertebral compression fractures refractory to nonoperative medical therapy in select patient populations.(1)(2)(3)(4)(5)(7) **(EG 2)**

For malignancy-related vertebral compression fractures, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A systematic review including 87 studies (3426 patients) evaluated the effectiveness and safety of vertebral augmentation (vertebroplasty or kyphoplasty) for malignant vertebral compression fractures and found that vertebral augmentation was associated with clinically relevant improvement in pain (measured by visual analog scale), disability (measured by Oswestry Disability Index), and Karnofsky Performance Status score, with a low rate of symptomatic procedure-related complications.(32) **(EG 1)** A systematic review including 111 studies (4235 patients) evaluated the effectiveness of vertebroplasty or kyphoplasty in patients with vertebral compression fractures due to spinal metastatic cancers, multiple myeloma, or hemangiomas and found that vertebroplasty and kyphoplasty were associated with a rapid reduction in pain intensity in patients refractory to conservative care (bed rest and opioids), decreased opioid pain medication use, and decreased functional disabilities. The author noted that the majority of the evidence was derived from observational studies and the clinical significance of these results in patients with advanced cancer is uncertain.(33)(34) **(EG 1)** Multiple specialty societies recommend percutaneous vertebral augmentation (vertebroplasty or kyphoplasty) for the treatment of painful vertebral compression fractures related to malignancy refractory to nonoperative medical therapy.(1)(2)(3)(4)(5)(6) **(EG 2)**

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment across patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 422.101(44); 42 CFR 419.22(45)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(46); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(47); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(48)

References

1. Barr JD, et al. Position statement on percutaneous vertebral augmentation: a consensus statement developed by the Society of Interventional Radiology (SIR), American Association of Neurological Surgeons (AANS) and the Congress of Neurological Surgeons (CNS), American College of Radiology (ACR), American Society of Neuroradiology (ASNR), American Society of Spine Radiology (ASSR), Canadian Interventional Radiology Association (CIRA), and the Society of NeuroInterventional Surgery (SNIS). *Journal of Vascular and Interventional Radiology* 2014;25(2):171-181. DOI: 10.1016/j.jvir.2013.10.001. (Reaffirmed 2023 Jul) [Context Link 1, 2, 3, 4] View abstract...
2. Chandra RV, et al. Vertebral augmentation: report of the Standards and Guidelines Committee of the Society of NeuroInterventional Surgery. *Journal of NeuroInterventional Surgery* 2014;6(1):7-15. DOI: 10.1136/neurintsurg-2013-011012. [Context Link 1, 2, 3] View abstract...
3. Khan MA, et al. Management of Vertebral Compression Fractures. ACR Appropriateness Criteria [Internet] American College of Radiology (ACR). 2022 Accessed at: <https://www.acr.org/>. [created 2010; accessed 2023 Oct 03] [Context Link 1, 2, 3, 4]
4. ACR-ASNR-ASSR-SIR-SNIS Practice Parameter for the Performance of Vertebral Augmentation. Resolution 19 [Internet] American College of Radiology. 2022 Accessed at: <https://www.acr.org/>. [created 2000; accessed 2023 Aug 15] [Context Link 1, 2, 3, 4]
5. Tsoumakidou G, et al. CIRSE guidelines on percutaneous vertebral augmentation. *Cardiovascular and Interventional Radiology* 2017;40(3):331-42. DOI: 10.1007/s00270-017-1574-8. (Reaffirmed 2023 Jun) [Context Link 1, 2, 3] View abstract...
6. Kyriakou C, et al. The role of cement augmentation with percutaneous vertebroplasty and balloon kyphoplasty for the treatment of vertebral compression fractures in multiple myeloma: a consensus statement from the International Myeloma Working Group (IMWG). *Blood Cancer Journal* 2019;9(3):Online. DOI: 10.1038/s41408-019-0187-7. [Context Link 1, 2] View abstract...
7. Hirsch JA, et al. Management of vertebral fragility fractures: a clinical care pathway developed by a multispecialty panel using the RAND/UCLA Appropriateness Method. *Spine Journal* 2018;18(11):2152-2161. DOI: 10.1016/j.spinee.2018.07.025. [Context Link 1, 2] View abstract...
8. Boss S, Srivastava V, Anitescu M. Vertebroplasty and kyphoplasty. *Physical Medicine and Rehabilitation Clinics* 2022;33(2):425-453. DOI: 10.1016/j.pmr.2022.01.008. [Context Link 1, 2, 3] View abstract...
9. Khan M, Kushchayev SV. Percutaneous vertebral body augmentations: the state of art. *Neuroimaging Clinics of North America* 2019;29(4):495-513. DOI: 10.1016/j.nic.2019.07.002. [Context Link 1, 2] View abstract...
10. Bousson V, Hamze B, Odri G, Funck-Brentano T, Orcel P, Laredo JD. Percutaneous vertebral augmentation techniques in osteoporotic and traumatic fractures. *Seminars in Interventional Radiology* 2018;35(4):309-323. DOI: 10.1055/s-0038-1673639. [Context Link 1] View abstract...
11. Chou R, Fu R, Dana T, Pappas M, Hart E, Mauer KM. Interventional Treatments for Acute and Chronic Pain: Systematic Review. Comparative Effectiveness Review No. 247. AHRQ Pub. No. 21-EHC030 [Internet] Agency for Healthcare Research and Quality. 2021 Sep Accessed at: <https://www.effectivehealthcare.ahrq.gov/>. [accessed 2023 Oct 17] DOI: 10.23970/AHRQEPCCER247. [Context Link 1, 2] View abstract...
12. Halvachizadeh S, et al. Systematic review and meta-analysis of 3 treatment arms for vertebral compression fractures: a comparison of improvement in pain, adjacent-level fractures, and quality of life between vertebroplasty, kyphoplasty, and nonoperative management. *Journal of Bone and Joint Surgery Reviews* 2021;9(10):Online. DOI: 10.2106/JBJS.RVW.21.00045. [Context Link 1] View abstract...
13. Lainez Ramos-Bossini AJ, Lopez Zuniga D, Ruiz Santiago F. Percutaneous vertebroplasty versus conservative treatment and placebo in osteoporotic vertebral fractures: meta-analysis and critical review of the literature. *European Radiology* 2021;31(11):8542-8553. DOI: 10.1007/s00330-021-08018-1. [Context Link 1] View abstract...
14. Lin S, Cheng M, Liang W, Yuan X. Efficacy of vertebral augmentation in the treatment of osteoporotic vertebral compression fractures: a systematic review and meta-analysis. *Annals of Palliative Medicine* 2021;10(11):11767-11775. DOI: 10.21037/apm-21-2834. [Context Link 1] View abstract...
15. Hinde K, Maingard J, Hirsch JA, Phan K, Asadi H, Chandra RV. Mortality outcomes of vertebral augmentation (vertebroplasty and/or balloon kyphoplasty) for osteoporotic vertebral compression fractures: a systematic review and meta-analysis. *Radiology* 2020;295(1):96-103. DOI: 10.1148/radiol.2020191294. [Context Link 1] View abstract...
16. Shankar JJS, Merdad R, Finitis S, Parker R. Imaging improves efficacy of vertebroplasty - a systematic review and meta-analysis. *Canadian Journal of Neurological Sciences* 2019;46(5):540-549. DOI: 10.1017/cjn.2019.236. [Context Link 1] View abstract...
17. Firanesco CE, et al. Vertebroplasty versus sham procedure for painful acute osteoporotic vertebral compression fractures (VERTOS IV): randomised sham controlled clinical trial. *British Medical Journal* 2018;361:k1551. [Context Link 1] View abstract...
18. Hansen EJ, Simony A, Carreon L, et al. Vertebroplasty vs. SHAM for treating osteoporotic vertebral compression fractures: a double blind RCT (VOPE). *Integrative Journal of Orthopaedics and Traumatology* 2019;2(4):1-6. [Context Link 1]
19. Yang EZ, et al. Percutaneous vertebroplasty versus conservative treatment in aged patients with acute osteoporotic vertebral compression fractures: a prospective randomized controlled clinical study. *Spine* 2016;41(8):653-60. DOI: 10.1097/BRS.0000000000001298. [Context Link 1] View abstract...
20. Clark W, et al. Safety and efficacy of vertebroplasty for acute painful osteoporotic fractures (VAPOUR): a multicentre, randomised, double-blind, placebo-controlled trial. *Lancet* 2016;388(10052):1408-16. DOI: 10.1016/S0140-6736(16)31341-1. [Context Link 1] View abstract...
21. Farrokhi MR, Alibai E, Maghami Z. Randomized controlled trial of percutaneous vertebroplasty versus optimal medical management for the relief of pain and disability in acute osteoporotic vertebral compression fractures. *Journal of Neurosurgery: Spine* 2011;14(5):561-9. DOI: 10.3171/2010.12.SPINE10286. [Context Link 1] View abstract...
22. Klazen CA, et al. Vertebroplasty versus conservative treatment in acute osteoporotic vertebral compression fractures (Vertos II): an open-label randomised trial. *Lancet* 2010;376(9746):1085-92. DOI: 10.1016/S0140-6736(10)60954-3. [Context Link 1] View abstract...
23. Rousing R, Andersen MO, Jespersen SM, Thomsen K, Lauritsen J. Percutaneous vertebroplasty compared to conservative treatment in patients with painful acute or subacute osteoporotic vertebral fractures: three-months follow-up in a clinical randomized study. *Spine* 2009;34(13):1349-54.

DOI: 10.1097/BRS.0b013e3181a4e628. [Context Link 1] View abstract...

24. Rousing R, Hansen KL, Andersen MO, Jespersen SM, Thomsen K, Lauritsen JM. Twelve-months follow-up in forty-nine patients with acute/semiacute osteoporotic vertebral fractures treated conservatively or with percutaneous vertebroplasty: a clinical randomized study. *Spine* 2010;35(5):478-82. DOI: 10.1097/BRS.0b013e3181b71bd1. [Context Link 1] View abstract...
25. Kallmes DF, et al. A randomized trial of vertebroplasty for osteoporotic spinal fractures. *New England Journal of Medicine* 2009;361(6):569-79. [Context Link 1] View abstract...
26. Buchbinder R, et al. A randomized trial of vertebroplasty for painful osteoporotic vertebral fractures. *New England Journal of Medicine* 2009;361(6):557-59. [Context Link 1] View abstract...
27. Wardlaw D, et al. Efficacy and safety of balloon kyphoplasty compared with non-surgical care for vertebral compression fracture (FREE): a randomised controlled trial. *Lancet* 2009;373(9668):1016-24. DOI: 10.1016/S0140-6736(09)60010-6. [Context Link 1] View abstract...
28. Beall DP, et al. Prospective and multicenter evaluation of outcomes for quality of life and activities of daily living for balloon kyphoplasty in the treatment of vertebral compression fractures: the EVOLVE trial. *Neurosurgery* 2019;84(1):169-178. DOI: 10.1093/neuros/nyy017. [Context Link 1] View abstract...
29. Percutaneous Vertebroplasty and Percutaneous Balloon Kyphoplasty for Treating Osteoporotic Vertebral Compression Fractures. NICE Technology Appraisal TA279 [Internet] National Institute for Health and Care Excellence. 2013 Apr (NICE reviewed 2016) Accessed at: <https://www.nice.org.uk/guidance/>. [accessed 2023 Aug 08] [Context Link 1]
30. Esses SI, et al. The treatment of symptomatic osteoporotic spinal compression fractures. *Journal of the American Academy of Orthopedic Surgeons* 2011;19(3):176-182. [Context Link 1] View abstract...
31. Eseonu KC, Panchmatia JR, Streetly MJ, Grauer JN, Fakouri B. The role of Vertebral Augmentation Procedures in the management of vertebral compression fractures secondary to multiple myeloma. *Hematological Oncology* 2023;41(3):323-334. DOI: 10.1002/hon.3102. [Context Link 1] View abstract...
32. Sorensen ST, Kirkegaard AO, Carreon L, Rousing R, Andersen MO. Vertebroplasty or kyphoplasty as palliative treatment for cancer-related vertebral compression fractures: a systematic review. *Spine Journal* 2019;19(6):1067-1075. DOI: 10.1016/j.spinee.2019.02.012. [Context Link 1] View abstract...
33. Health Quality Ontario. Vertebral augmentation involving vertebroplasty or kyphoplasty for cancer-related vertebral compression fractures: a systematic review. *Ontario Health Technology Assessment Series* 2016;16(11):1-202. [Context Link 1] View abstract...
34. Berenson J, et al. Balloon kyphoplasty versus non-surgical fracture management for treatment of painful vertebral body compression fractures in patients with cancer: a multicentre, randomised controlled trial. *Lancet Oncology* 2011;12(3):225-35. DOI: 10.1016/S1470-2045(11)70008-0. [Context Link 1] View abstract...
35. Lutz S, et al. Palliative radiation therapy for bone metastases: update of an ASTRO evidence-based guideline. *Practical Radiation Oncology* 2017 Jan - Feb;7(1):4-12. DOI: 10.1016/j.prro.2016.08.001. (Reaffirmed 2023 Jul) [Context Link 1] View abstract...
36. Tseng CL, et al. Spine stereotactic body radiotherapy: indications, outcomes, and points of caution. *Global Spine Journal* 2017;7(2):179-97. DOI: 10.1177/2192568217694016. [Context Link 1] View abstract...
37. Sindt JE, Brogan SE. Interventional treatments of cancer pain. *Anesthesiology Clinics* 2016;34(2):317-339. DOI: 10.1016/j.andclin.2016.01.004. [Context Link 1] View abstract...
38. Nonopioid drugs for pain. *Medical Letter on Drugs and Therapeutics* 2018;60(1540):25-32. [Context Link 1, 2] View abstract...
39. Genev IK, Tobin MK, Zaidi SP, Khan SR, Amirouche FML, Mehta AI. Spinal compression fracture management: a review of current treatment strategies and possible future avenues. *Global Spine Journal* 2017;7(1):71-82. DOI: 10.1055/s-0036-1583288. [Context Link 1] View abstract...
40. McCarthy J, Davis A. Diagnosis and management of vertebral compression fractures. *American Family Physician* 2016;94(1):44-50. [Context Link 1] View abstract...
41. Descamps J, Lamerain M, Chenguel Z, Jubert P, Rousseau MA. Vertebral compression fractures treated in acute by instrumented kyphoplasty: early and mid-term clinical and radiological results. *BioMed Research International* 2019;2019:1386510. DOI: 10.1155/2019/1386510. [Context Link 1] View abstract...
42. Chabert E, et al. Vertebroplasty versus bracing in acute vertebral compression fractures: A prospective randomized trial. *Annals of Physical and Rehabilitation Medicine* 2023;66(6):Online. DOI: 10.1016/j.rehab.2023.101746. [Context Link 1] View abstract...
43. Ferreira ML, March L. Vertebral fragility fractures - How to treat them? Best Practice and Research. *Clinical Rheumatology* 2019;33(2):227-235. DOI: 10.1016/j.berh.2019.03.017. [Context Link 1] View abstract...
44. Centers for Medicare and Medicaid Services. "Requirements relating to basic benefits." 42 CFR 422.101 Washington, DC 2023 Jun 05 [accessed 2023 Oct 30] Accessed at: <https://www.ecfr.gov/>. [Context Link 1]
45. Centers for Medicare and Medicaid Services. "Hospital services excluded from payment under the hospital outpatient prospective payment system." 42 CFR 419.22 Washington, DC 2023 Jul [accessed 2023 Oct 30] Accessed at: <https://www.ecfr.gov/>. [Context Link 1]
46. Centers for Medicare and Medicaid Services. Medicare Benefit Policy Manual. Chapter 14 - Medical Devices Rev. 198 [Internet] Centers for Medicare and Medicaid Services. 2014 Nov 06 Accessed at: <https://www.cms.gov/Regulations-and-Guidance/Guidance/Manuals/>. [accessed 2023 Oct 02] [Context Link 1]
47. Centers for Medicare and Medicaid Services. Medicare Benefit Policy Manual. Chapter 15 - Covered Medical and Other Health Services Rev. 12171 [Internet] Centers for Medicare and Medicaid Services. 2023 Aug 03 Accessed at: <https://www.cms.gov/Regulations-and-Guidance/Guidance/Manuals/>. [accessed 2023 Oct 25] [Context Link 1]
48. Centers for Medicare and Medicaid Services. Medicare Benefit Policy Manual. Chapter 16 - General Exclusions From Coverage Rev. 198 [Internet] Centers for Medicare and Medicaid Services. 2014 Nov 06 Accessed at: <https://www.cms.gov/Regulations-and-Guidance/Guidance/Manuals/>. [accessed 2023 Oct 02] [Context Link 1]

49. Medicare Coverage Database. [Internet] Centers for Medicare and Medicaid Services. Accessed at: <https://www.cms.gov/medicare-coverage-database/search.aspx>? Updated 2023 [accessed 2023 Oct 20] [Context Link 1]

Codes

CPT®: 22510, 22511, 22512, 22513, 22514, 22515

HCPCS: C7504, C7505, C7507, C7508

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